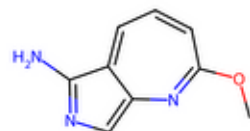


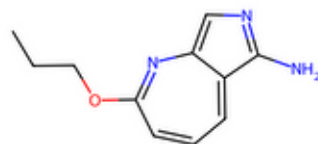
Molecular evolution at key LDM (Direct-Softmax) turning points

A r1 Anchor core



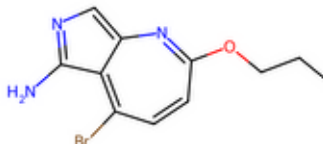
dHV 8.47 | Vina -6.53 | NN 6.30
MW 175 | LogP 1.2 | F 0 | N 3
LLM intent: 8-methoxy local edit

B r7 Propoxy homologation



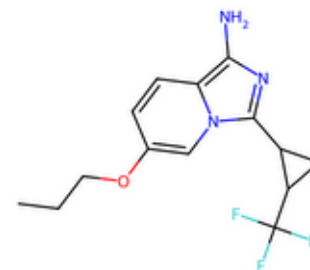
dHV +2.09 | Vina -6.95 | NN 6.57
MW 203 | LogP 2.0 | F 0 | N 3
LLM intent: propoxy chain variation

C r13 Br consensus (17 sources)



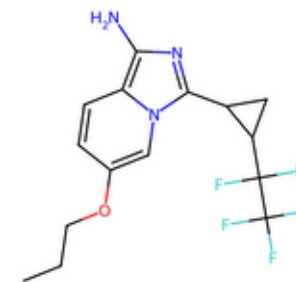
dHV +2.12 | Vina -7.11 | NN 6.93
MW 282 | LogP 2.7 | F 0 | N 3
LLM intent: bromo substitution

D r19 CF3-cyclopropyl branch



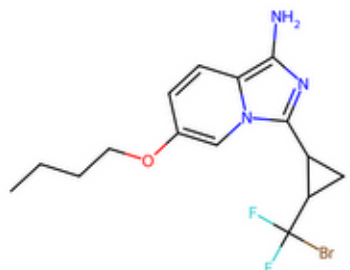
dHV +2.41 | Vina -7.50 | NN 7.40
MW 299 | LogP 3.4 | F 3 | N 3
LLM intent: trifluoromethyl cyclopropyl exploration

E r21 C2F5 extension



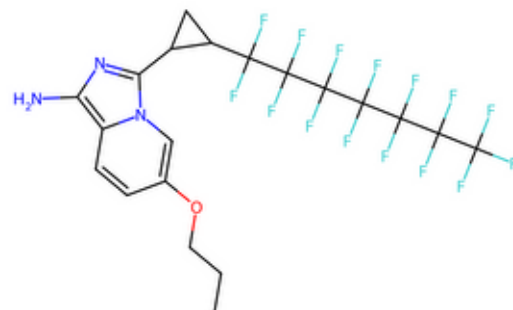
dHV +3.18 | Vina -8.73 | NN 7.43
MW 349 | LogP 4.0 | F 5 | N 3
LLM intent: extend fluorinated chain

F r26 Butoxy-Br activity jump



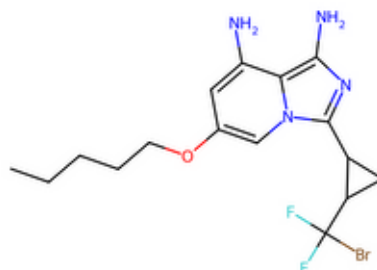
dHV +3.90 | Vina -7.82 | NN 8.04
MW 374 | LogP 4.2 | F 2 | N 3
LLM intent: increase alkoxy, vary halogen

G r29 Long perfluoro tail



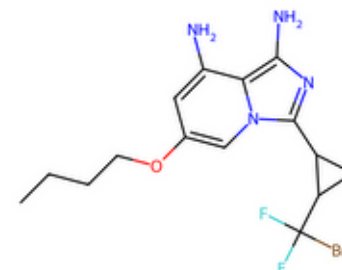
dHV +0.89 | Vina -10.32 | NN 7.06
MW 599 | LogP 7.2 | F 15 | N 3
LLM intent: Longer perfluoroalkyl chain

H r46 Extra aryl NH2



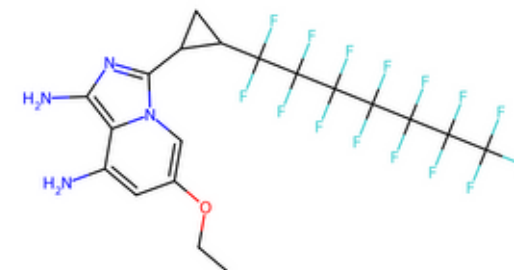
dHV +1.60 | Vina -8.02 | NN 8.23
MW 403 | LogP 4.2 | F 2 | N 4
LLM intent: add amino for H-bond

I r49 Activity branch refined



dHV +0.85 | Vina -8.29 | NN 8.33
MW 389 | LogP 3.8 | F 2 | N 4
LLM intent: add amino to ring, shorten chain

J r51 Elite crossover



dHV +0.30 | Vina -10.02 | NN 7.69
MW 600 | LogP 6.4 | F 15 | N 4
LLM intent: cross NH2 onto long fluorinated scaffold